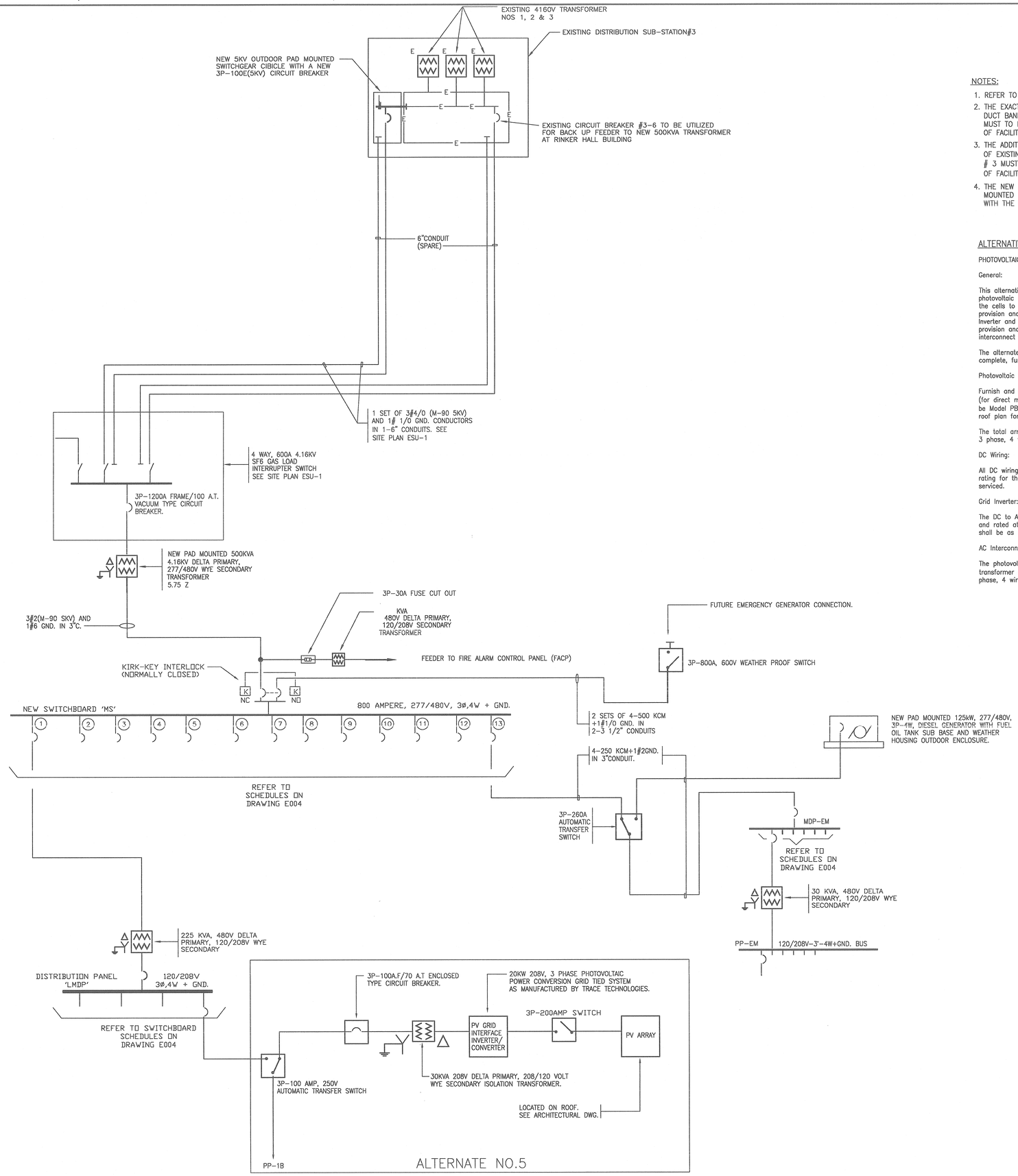

NOTES:

1. REFER TO DWG. E001 FOR SYMBOL LIST AND ABBREVIATIONS.
2. THE EXACT LOCATION AND ROUTING OF NEW ELECTRIC CONDUIT DUCT BANK FROM EXISTING SUBSTATION # 3 TO NEW BUILDING MUST TO BE COORDINATED WITH UNIVERSITY OF FLORIDA DEPT OF FACILITIES.
3. THE ADDITION OF NEW 5KV CIRCUIT BREAKER # 3-10 AND USE OF EXISTING CIRCUIT BREAKER # 3-6 AT EXISTING SUBSTATION # 3 MUST BE COORDINATED WITH UNIVERSITY OF FLORIDA DEPT OF FACILITIES.
4. THE NEW CIRCUIT BREAKER, 4-WAY SF6 SELECTOR SWITCH, PAD MOUNTED TRANSFORMER AND CONDUIT DUCT BANKS MUST COMPLY WITH THE UNIVERSITY OF FLORIDA FACILITY REQUIREMENTS.

ALTERNATIVE NO.5

PHOTOVOLTAIC SYSTEM

General:

This alternative covers the complete installation of a roof mounted photovoltaic system, including the photovoltaic array, the wiring of the cells to produce DC current input to a Grid Inverter, the provision and installation of the Inverter, AC wiring between the Inverter and the installed electrical power distribution system, and the provision and installation of a protective interface device at the interconnect point.

The alternate price should include all labor and material for a complete, functional installation.

Photovoltaic Collectors:

Furnish and install 2,000 square feet of Solarex SX Series Modules (for direct mount on the flat roof surface). Collector panels shall be Model PB/SX-65D, rated at 65 watts output. See architectural roof plan for location.

The total array will be rated for a peak output of 20 kw, 208 volt, 3 phase, 4 wire.

DC Wiring:

All DC wiring shall be sized at 1.5 times the nominal peak current rating for the photovoltaic array or section of that array being serviced.

Grid Inverter:

The DC to AC inverter will be of grid connection quality and stability, and rated at 20 kw, 208 volt, 3 phase, 4 wire. The grid inverter shall be as manufactured by Trace Tech.

AC Interconnect Device:

The photovoltaic array output will be routed to Grid inverter, isolation transformer and then connected to the building 120/208 volt, 3 phase, 4 wire power system at location indicated on electrical plan.

PROJECT NORTH

M.E. RINKER, Sr. HALL
SCHOOL OF BUILDING CONSTRUCTION

UNIVERSITY OF FLORIDA-GAINESVILLE
PROJECT NO. BR-191

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